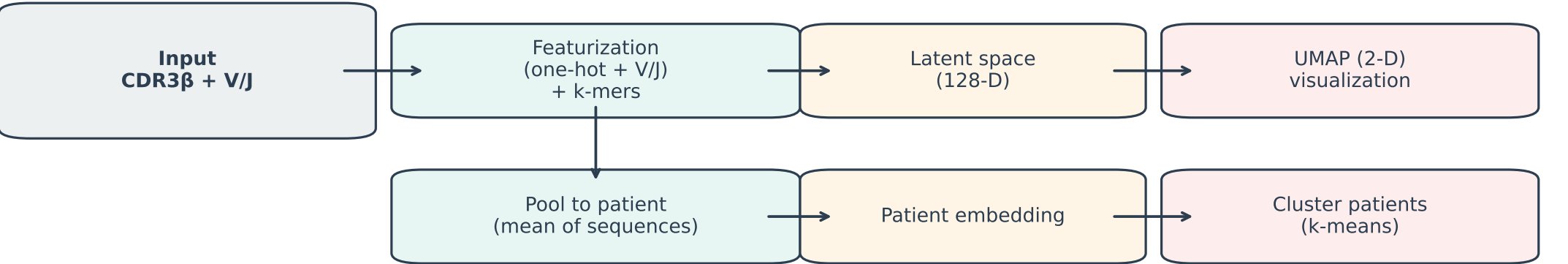


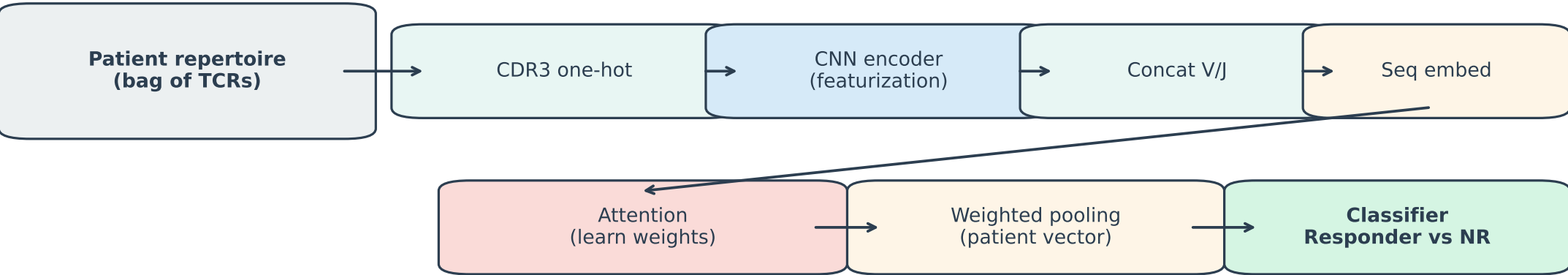
Figure S12: Featurization schematics for unsupervised and supervised analyses

A. Unsupervised representation + UMAP (no labels used for embedding)



Note: This is an unsupervised structure/visualization tool.
It does not assign 'Responder/Non-responder' without labeled outcomes.

B. Supervised DeepTCR MIL (CNN featurization + attention → patient prediction)



The CNN featurization + attention layer identifies which clonotypes contribute most to the patient-level prediction.